TERMINAL THREE-FOLD SINGULARITIES IN POSITIVE CHARACTERISTIC NEED NOT BE COHEN–MACAULAY

ARGUS OPEN RESEARCH PROJECT

Abstract. In characteristic zero, terminal singularities are rational and hence Cohen–Macaulay by Elkik and Kollár–Mori. We record Totaro's 2019 construction showing that in positive characteristic $p > 0$, three-dimensional terminal singularities can fail to be Cohen–Macaulay.

1. Main Refutation

Theorem 1.1 (Totaro, 2019)**.** *For every prime $p > 0$, there exist three-dimensional terminal singularities over a field of characteristic p that are not Cohen–Macaulay (constructed as quotients of smooth varieties by actions of $\mathbb{Z}/p\mathbb{Z}$ or finite group schemes with controlled discrepancy).*

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